

A192 – Gauge Sector: Torus Topology and Covariant Derivatives

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.1 Purpose

[STIPULATION] A190 names the gauge sector as the largest remaining gap. This document carries out the derivation: from torus topology to the covariant derivative and the gauge Lagrangian, for all three factors of the SM gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$.

.2 Starting Point: Torus and Flux Quantisation

[B] On T^4 there exist non-contractible loops. The flux of a $U(1)$ connection through such a loop C is quantised: every continuous deformation of C leaves the flux invariant. This is the basic structure of all three gauge groups.

.3 $U(1)_Y$: Electromagnetic Charge

[B] The wave function ψ accumulates a phase upon traversing the torus loop C :

$$\psi \longrightarrow \psi \cdot \exp\left(i \frac{e}{\hbar} \Phi\right), \quad \Phi = \oint_C A_\mu dx^\mu = n \frac{h}{e}, \quad n \in \mathbb{Z}.$$

The monodromy $\exp(i \cdot 2\pi n) = 1$ is satisfied for all $n \in \mathbb{Z}$. This is the $U(1)$ gauge structure: the phase $e^{i\alpha(x)}$ with $\alpha(x) = (e/\hbar) \int A_\mu dx^\mu$.

[B] The covariant derivative and the field-strength tensor follow:

$$D_\mu = \partial_\mu + ieA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

The gauge Lagrangian follows from minimal coupling: $F_{\mu\nu}$ is the curvature of the connection A_μ , $\mathcal{L}_{\text{EM}}$ is its norm. No free step.

.4 $SU(3)_C$: Colour Charge from Linking Number

[B] Three flux tubes on $T^4/\mathbb{Z}_3$ carry linking numbers $\ell k(i, j) \in \mathbb{Z}$ for $i, j \in \{1, 2, 3\}$ (tubes i and j). The independent non-trivial states are:

Linking state	Generator	Physics
$\ell k(1, 2) = \pm 1, \text{ rest } 0$	λ_1, λ_2	Gluon $r \leftrightarrow g$
$\ell k(1, 3) = \pm 1, \text{ rest } 0$	λ_4, λ_5	Gluon $r \leftrightarrow b$
$\ell k(2, 3) = \pm 1, \text{ rest } 0$	λ_6, λ_7	Gluon $g \leftrightarrow b$
$\ell k(1, 1) - \ell k(2, 2)$	λ_3	Gluon neutral (1)
$\ell k(1, 1) + \ell k(2, 2) - 2 \ell k(3, 3)$	λ_8	Gluon neutral (2)

[B] Count: $\binom{3}{2} = 3$ pairs, each with real and imaginary part, gives 6 off-diagonal; plus 2 traceless diagonal generators (rank-1 reduction of the 3×3 tracelessness condition) gives $6 + 2 = 8$ independent generators – exactly $\dim(\mathfrak{su}(3))$. The gauge group $SU(3)_C$ is topologically forced, not postulated.

[B] Covariant derivative and Lagrangian analogous to $U(1)$:

$$D_\mu = \partial_\mu + i g_s G_\mu^a \lambda_a / 2, \quad G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c,$$

$$\mathcal{L}_{\text{strong}} = -\frac{1}{4} G_{\mu\nu}^a G^{a,\mu\nu}.$$

The non-commutativity $f^{abc} \neq 0$ follows from the non-commutativity of the linking-number operations for three tubes.

.5 $SU(2)_L$: Weak Isospin Group from Chiral Projection

[B] The chiral projection P_+ (A095) selects the $m \geq 0$ sector of the torus. The spinor-doublet character of the weak interaction follows from the spin- $\frac{1}{2}$ structure (winding $(1, 1)$, A263): the two components of the isospin doublet (u_L, d_L) and (ν_L, e_L) respectively are the two independent $m = 0$ and $m = +1$ sectors after projection.

[B] The three generators τ^+, τ^-, τ^3 (Pauli matrices) of $SU(2)_L$ correspond to the three independent transitions between the two projected states. Covariant derivative:

$$D_\mu = \partial_\mu + i g_w W_\mu^a \sigma_a / 2, \quad W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g_w \epsilon^{abc} W_\mu^b W_\nu^c,$$

$$\mathcal{L}_{\text{weak}} = -\frac{1}{4} W_{\mu\nu}^a W^{a,\mu\nu}.$$

.6 Overall Picture: Gauge Lagrangian from Torus Topology

[B] The complete Standard Model gauge Lagrangian:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G_{\mu\nu}^a G^{a,\mu\nu} - \frac{1}{4} W_{\mu\nu}^a W^{a,\mu\nu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

follows from three topological structures of $T^4/\mathbb{Z}_3$:

- $U(1)_Y$: flux quantisation through non-contractible loop
- $SU(3)_C$: linking number of three $\mathbb{Z}_3$ flux tubes ($6 + 2 = 8$ generators)
- $SU(2)_L$: chiral projection of the spin- $\frac{1}{2}$ sector (A095)

Group	Topological origin	Generators	Gauge bosons
$U(1)_Y$	Flux quantisation	1	Photon (γ)
$SU(2)_L$	Chiral projection	3	W^+, W^-, Z^0
$SU(3)_C$	Linking number	8	8 gluons
Total		12	12 SM gauge bosons

.7 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
$\ell k(i, j)$	Linking number of tubes i and j on $T^4/\mathbb{Z}_3$
λ_a	Gell-Mann matrices, generators of $SU(3)$
σ_a	Pauli matrices, generators of $SU(2)$
G_μ^a, W_μ^a, A_μ	Gauge fields of $SU(3)$, $SU(2)$, $U(1)$
f^{abc}	Structure constants of $SU(3)$
ε^{abc}	Levi-Civita symbol, structure constants of $SU(2)$
g_s, g_w, e	Coupling constants (not derived from ξ)

.8 Verification Script

- Count of linking states; Gell-Mann matrix identification; tracelessness condition; $6+2 = 8$:
python/A_Serie_Skripte/a192_eichsektor.py

.9 Open Questions

The coupling constants g_s, g_w, e are not derived from ξ . The group structure follows from the topology; the strength of the couplings does not. This is the remaining gap.

The running of the couplings $(g_s(\mu), g_w(\mu))$ as functions of the energy scale μ is not derived from FFGFT.

The Higgs mechanism – the spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$ – is not derived from the torus geometry. The Higgs-EFT consistency check (A190, 2.3 %) is a hint, not a proof.

.10 Supersedes

This document systematises the gauge-sector content of the old corpus (charge quantisation and colour charge from torus topology, Lagrangian structure) self-contained without old-corpus citations. It supplements A190 (SM as projection), A095 (chiral projection) and A075 (field theory).

.11 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.